

Text Editors for Programmers

vi Editor

The vi editor (short for visual editor) is a screen editor which is available on almost all Unix systems.

Once you have learned vi, you will find that it is a fast and powerful editor.

vi has no menus but instead uses combinations of keystrokes in order to accomplish commands.

emacs



*People sometimes ask me if it is a sin in the Church of Emacs to use vi.
Using a free version of vi is not a sin; it is a penance. So happy hacking.
(Richard Stallman)*



- Powerfull and efficient
- Highly configurable
- Program your text!

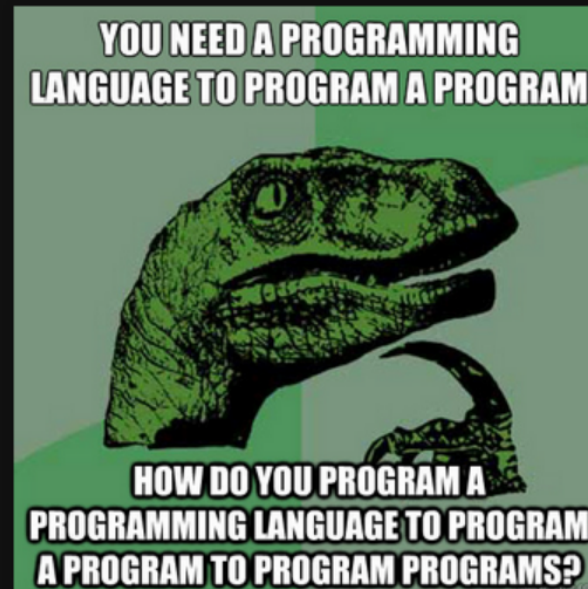
Often called a *programmer's* editor,
Vim is a REAL Jedi Knight text editor!

{ Exercise }

vi intro - **the cheat sheet method**

Introduction to Computer Programming

Programming Languages



Programming Languages

Languages designed to communicate instructions to the machine

Machine Language - B0 61

Assembly Language - MOV AL 61

C - int employeeCount = 61;

Java - int employeeCount = 61;

JavaScript - let employeeCount = 61;

Increasingly closer to...

Natural Language - The number of employees is 61

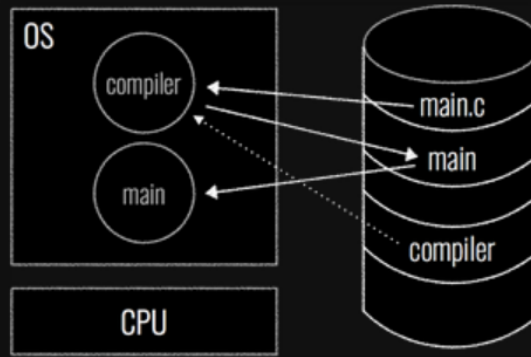
Compiled vs Interpreted

A compiled language is one where the program, once compiled, is expressed in the instructions of the target machine

An interpreted language is one where the instructions are not directly executed by the target machine, but instead read and executed by some other program

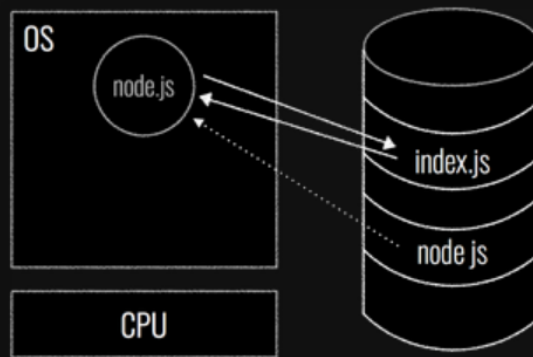
Compilers generate native machine code, interpreters read source code and generate machine code on the fly using some sort of run-time system

Compiled Code



- the main.c file contains the C source code to be compiled
- compiler is a binary program that compiles C source code
- main is the compiler output, an executable binary resulting from compiling the source code

Interpreted Code



- the index.js file contains the JavaScript source code to be interpreted
- node.js is a binary program that interprets JavaScript source code

{ Hands On }

Poliglot Hello World

Algorithms

In simple terms, an algorithm is a set of steps to accomplish a task

{ Board Exercise }

Algorithm Pseudocode

Searching Algorithms

Linear Search



The magic number we are searching for is 281. Let's try the first one!

Searching Algorithms

Linear Search

- numbers do not need to be sorted
- search time increases with number of elements

{ Board Exercise }

Pseudocode for Linear Search

Searching Algorithms

Binary Search



The magic number we are searching for is 377. Let's try the first one!

{ Board Exercise }

Pseudocode for Binary Search

Sorting Algorithms

Selection Sort

Repeatedly selects the next-smallest element and swaps it into place

{ Board Exercise }

Pseudocode for Selection Sort

Sorting Algorithms

Insertion Sort

Repeatedly selects a new value and inserts it into a sorted sub-array

{ Board Exercise }

Pseudocode for Insertion Sort

Asymptotic Notation

How to measure the speed of an algorithm?

big O notation measures how fast a program run time grows as the size of the input increases towards infinity

asymptotic notation does not depend on hardware performance

linear search is $O(n)$, as the time it takes to find element grows linear to the number of elements

string length is $O(n)$ if we need to count number of chars in string one by one

string length is $O(1)$ if length is stored before storing the string.

binary search is $O(\log n)$ as for each time we double the number of input elements we need one more attempt to find the element

How about our algorithms?

Linear Search

$O(n)$ time complexity

Binary Search

$O(\log n)$ time complexity

Selection Sort

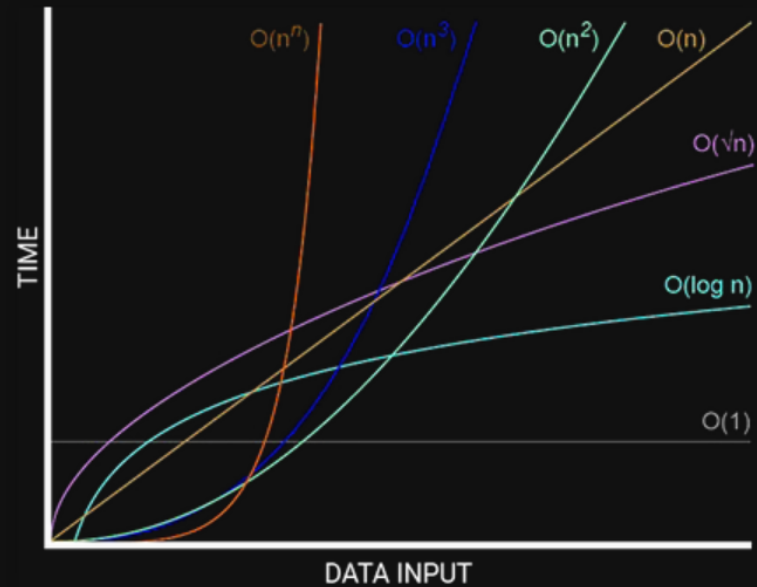
$O(n^2)$ time complexity, inefficient in large lists

Insertion Sort

Worst Case - $O(n^2)$ time complexity

Adaptive - $O(n)$ time complexity when nearly sorted

Big-O Complexity



{ Hands On }

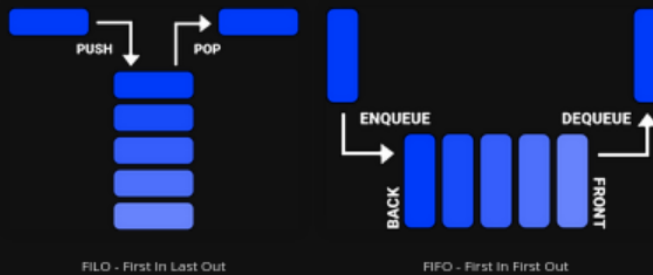
Visualize Different Sorting Algorithms in Action

Data Structures

Particular way of organizing and accessing data in a computer so that it can be used efficiently

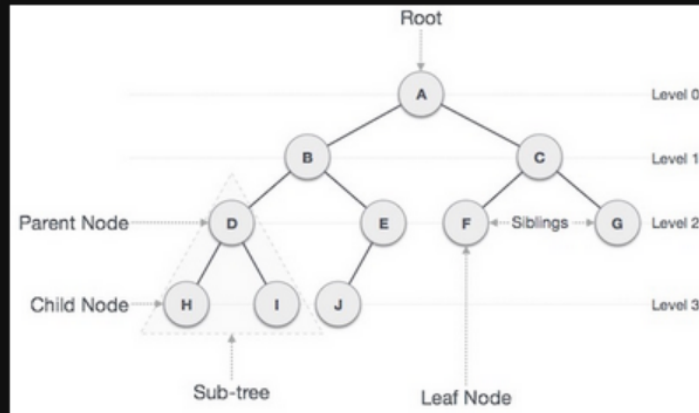
FILO and FIFO

a Stack and a Queue



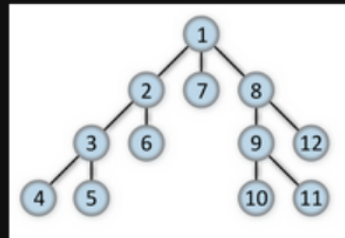
Tree

Hierarchical data structure used to store, manipulate and search information that naturally forms a hierarchy



Trees where each node can have a maximum of two children are called Binary Trees

Graph Traversal Algorithm



Find paths between a starting and a target node.

Depth First Search visits child nodes first, requires usage of a Stack

Breadth First Search visits neighbor nodes first, requires usage of a Queue

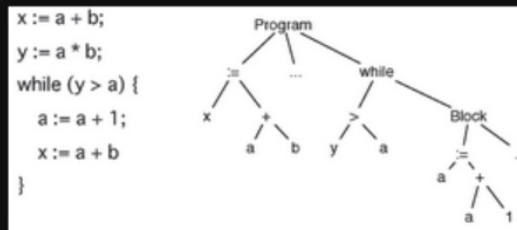
Example Use Case

Abstract Syntax Trees

AST is a tree representation of the abstract syntactic structure of source code written in a programming language

Each node of the tree denotes a construct occurring in the source code

Used by compilers, parsers, code generators, etc.

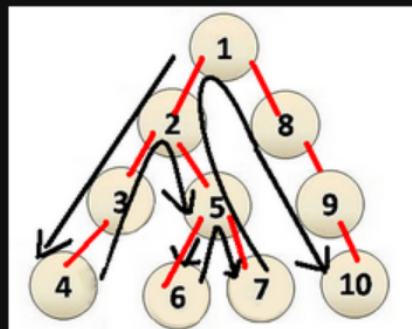


Another Use Case

Creating and solving Maze like puzzles



DFS



{ Board Exercise }

Pseudocode for generating a Maze

{ Board Exercise }

Pseudocode for solving a Maze

Recursion



1. Each recursive call should be on a smaller instance of the same problem, that is, a smaller subproblem.
2. The recursive calls must eventually reach a base case, which is solved without further recursion.

The Call Stack

A Stack data structure is utilized to store function execution context:

```
function setupGame() {  
  getPlayers();  
}  
  
function getPlayers() {  
  askP1Name();  
  askP2Name();  
}
```

askP2Name()

getPlayers()

setupGame()

{ Board Exercise }

Factorial Pseudocode

{ Board Exercise }

Iterative Palindrome Pseudocode

{ Board Exercise }

Recursive Palindrome Pseudocode

RegEx

Regular expressions define search patterns,
used to find occurrences in text

Used on command line utilities, Text Editors, IDEs
and in most Programming Languages

Some people, when confronted with a problem,
think "I know, I'll use regular expressions."

Now they have two problems.

Regex Flavours

Mostly compatible across programs and programming languages

Comes in different flavours:

- Basic Regular Expressions (grep)
- Extended Regular Expressions (egrep)
- Perl Compatible Regular Expressions (PCRE - Perl, PHP, Java)
- JavaScript Regular Expressions (subset of PCRE)
- Python Regular Expressions (similar to PCRE)

Exact Match

regular expression → **c k s**

ro**cks**.



match

is okay.

no match

Multiple Matches

regular expression



a	p	p	l	e
---	---	---	---	---

Scrapple from the apple.



match 1



match 2

Regex Metacharacters

Character	Meaning	Example
*	Match zero, one or more of the previous	Ah* matches A or Ahhhhhhhhhh
?	Match zero or one of the previous	Ah? matches A or Ah
+	Match one or more of the previous	Ah+ matches Ah or Ahhh but not A
\	Used to escape a special character	Hungry\? matches Hungry?
.	Wildcard, matches any character	do.* matches dog, door, dot
	Matches previous or next character/group	(Mon) (Tues)day matches Monday or Tuesday
()	Groups characters	See previous example
[]	Matches a range of characters	[0-9]+ matches any positive integer
{ }	Matches a specified number of occurrences	[0-9]{2,4} matches 12, 1234 but not 12345
^	Beginning of a string or a negation if within a []	^Hello matches string that begins with Hello [^0-9] matches any character not 0-9
\$	End of a string	ing\$ matches exciting but not ingenious

Popular Use Case

Form Validation

Contact Information

Business Name
At least 3 characters long

Address
min length 3

min length 3 min length 3 min length 3

Sri Lanka

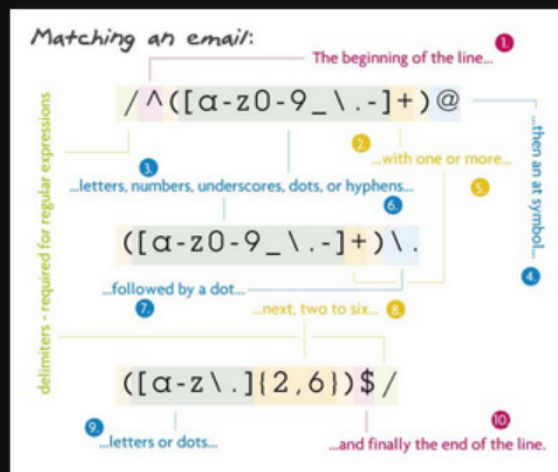
Phone
Your Telephone number must be at least 3 characters long

Fax
Your fax number must be at least 3 characters long

Business Email
Please enter a valid email address.

Website
Your web site must be at least 8 characters long

Email matching



{ Hands On }

Regex on the Command Line, Text Editor, Code

{ Board Exercise }

Regular expressions

{ Exercise }

Regex Tutorial